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Roll No

CE-3002-CBGS

B.E., III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Fluid Mechanics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain types of fluid using stress strain diagram. 7
b) Distinguish between
 - i) Real fluids and ideal fluids
 - ii) Specific weight and specific volume. 7
2. a) State and prove Pascal's law. 7
b) Explain stream function and velocity potential. 7
3. a) Discuss various arrangements of Pitot tube used in pipes. 7
b) A pipeline of length 2000 m is used for power transmission. If 110.3625 kW power is to be transmitted through the pipe in which water having a pressure of 490.5 N/cm^2 at inlet is flowing. Find the diameter of the pipe and efficiency of transmission if the pressure drop over the length of pipe is 98.1 N/cm^2 . Take $F = 0.0065$. 7

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4. a) A rectangular channel of width 4 m is having a bed slope of 1 in 1500. Find the discharge through the channel. Take $C = 50$. 7
- b) Find the discharge through a trapezoidal channel of width 8 m and side slope of 1 horizontal to 3 vertical. The depth of flow water is 2.4 m and value of Chezy's constant $C = 50$. The slope of the bed of the channel is given 1 in 4000. 7
5. a) Differentiate between.
- i) Impulse and reaction turbines 7
- ii) Radial and axial flow turbines 7
- b) What is drag and lift? Explain different type of drag an immersed body. 7
6. a) Find the displacement thickness, the momentum thickness and energy thickness for the velocity distribution in the boundary layer given by:
- $$\frac{u}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \quad 7$$
- b) Write a short note on "Boundary layer formation" over flat plate. 7
7. a) Derive Euler's equation for flow along streamline and reduce the Bernoulli's equation for the same. 7
- b) The stream function for a two dimensional flow is given by $\psi = 8xy$. Calculate the velocity at the point P(4,5). Find also velocity potential function ϕ . 7
8. Write short notes on : 14
- i) Total energy line
- ii) Camber
- iii) Weber Number
- iv) Reynolds Number